浙江大学

ZJUI Institute 2026 Summer Research Project Summary

Project title：Simulation-Based Robot Manipulation for Drone Battery Replacement

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Abstract

The operational tolerance of multi-rotor drones is significantly limited by battery capacity, requiring frequent manual interventions that disrupt autonomous missions. This research presents the design and simulation of two novel robotic end-effectors for automated drone battery replacement. Design A is a dual-battery end-effector that utilizes a compliant belt-driven mechanism to securely grasp the narrow, extended tip of the battery, followed by a 180-degree roll rotation to rapidly swap the depleted battery with a charged one without external positioning. Design B combines a parallel gripper with a suction cup. It is designed for batteries that sit deep inside the drone, leaving only a tiny exposed edge to grab. The suction cup initially pulls the battery out slightly, allowing the parallel fingers to securely grasp it, requiring the robot to place the discharged unit and pick up a new one. Both designs target industrial platforms such as the DJI Matrice 4 series. To bypass the complex and computationally expensive Inverse Kinematics (IK) calculations typically required for robotic manipulation, a simulation framework is developed in Isaac Lab using a 6-DOF UR10 manipulator with a parallel-jaw gripper. Through a learning-based approach, the robot autonomously learns the trajectory policy to effectively engage the grippers and execute the battery swap sequence. Simulation results demonstrate the feasibility of both mechanical designs and validate the learned control policies, establishing a foundation for subsequent physical implementation. The two mechanical gripper designs (belt-driven and suction-assisted) are presented as conceptual proposals, and the simulation validates the grasp-and-align idea that both designs would rely on.

1 Introduction

Unmanned Aerial Vehicles (UAVs) have become an important tools for infrastructure inspection, precision agriculture, and emergency response. The

majority of industrial multi-rotor systems, such as the DJI Matrice 4 series, offer limited flight times (around 49 minutes), requiring frequent manual battery swaps. Automated battery management systems have emerged as a solution. However, the critical bottleneck lies within the manipulation mechanism - the end-effector, that must reliably interact with the drone's power interface.

A significant challenge is that drone batteries often have narrow extended tips and are deeply positioned inside the drone body, leaving only a very small exposed part for gripping. This makes them difficult to grasp with rigid traditional fingers due to the small contact area and limited accessibility. The risk of damage during the swap operation demands a gripper that is precise, fast, and highly tolerant of alignment variations. This research explores two distinct end-effector designs:

- Design A (Dual-Battery Belt Gripper): A self-contained system that preloads a charged battery and uses a compliant belt-driven mechanism with a 180° roll to swap batteries in a single, continous motion.
- Design B (Suction-Assisted Parallel Gripper): An alternative that uses a central suction cup to pull the battery out of its slot, increasing the exposed area/length so that parallel fingers can securely grasp the battery. This design requires the robot to place the discharged battery at a specific location before picking up a charged one.

To avoid the significant computational overhead of solving complex Inverse Kinematics (IK) equations for these sophisticated gripping tasks, this research uses Isaac Lab with a UR10 manipulator. Instead of hard-coding conventional analytic IK, the simulation performs deep reinforcement learning (RL) to train the robot's control policy, allowing it to autonomously discover the optimal trajectory for approaching and grasping a battery-sized target.

2 Methodology

2.1 Design A: Dual-Battery Belt Gripper

The primary innovation of this design lies in its integrated dual-battery storage and caterpillar-belt type of fingers. The gripper consists of four parallel fingers (two pairs), each dedicated to holding a single battery, complemented by a fifth and sixth pair of "clamping" fingers.

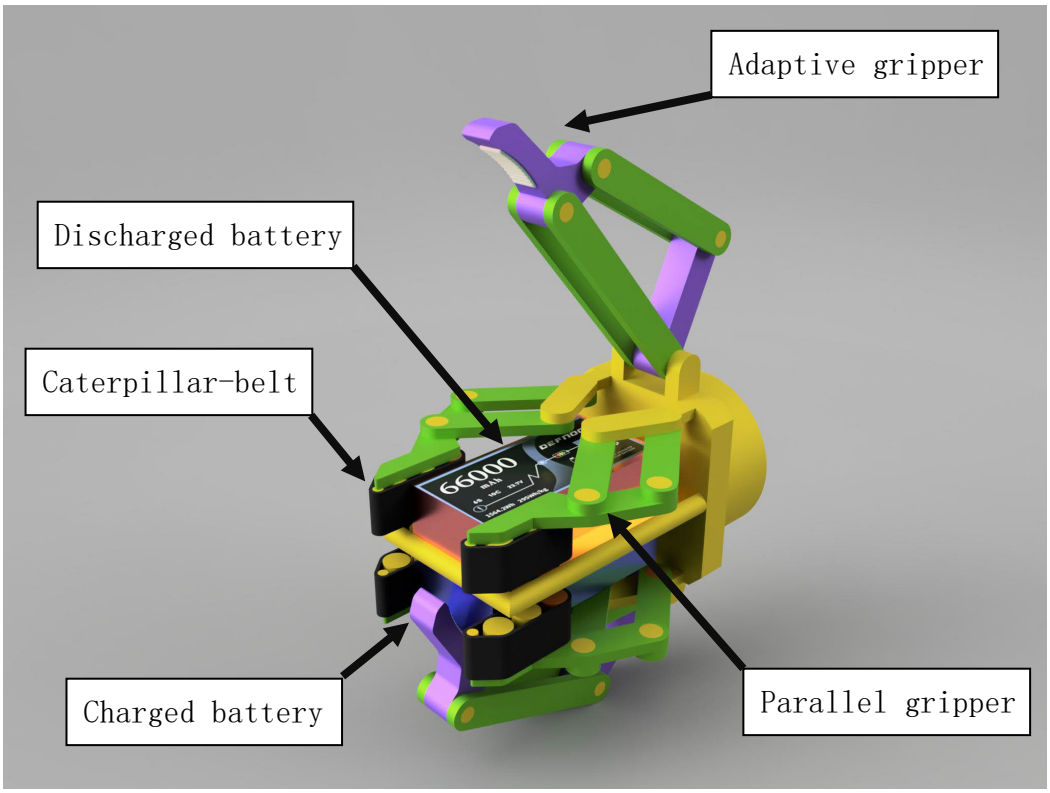


Fig. 1: Rendering of Design A (Dual-Battery Belt Gripper)

- Belt-Driven Fingers: The belt generates high friction and distributes force evenly across the narrow extended tip of the battery, preventing slipping and damage.
- Clamping Mechanism & 180° Roll: Once the battery is extracted, the upper fingers curl to lock it in place. The entire assembly then rotates 180 degrees on its roll axis, positioning the charged battery for insertion into the drone slot, eliminating the need for secondary grippers or external reorientation stations.

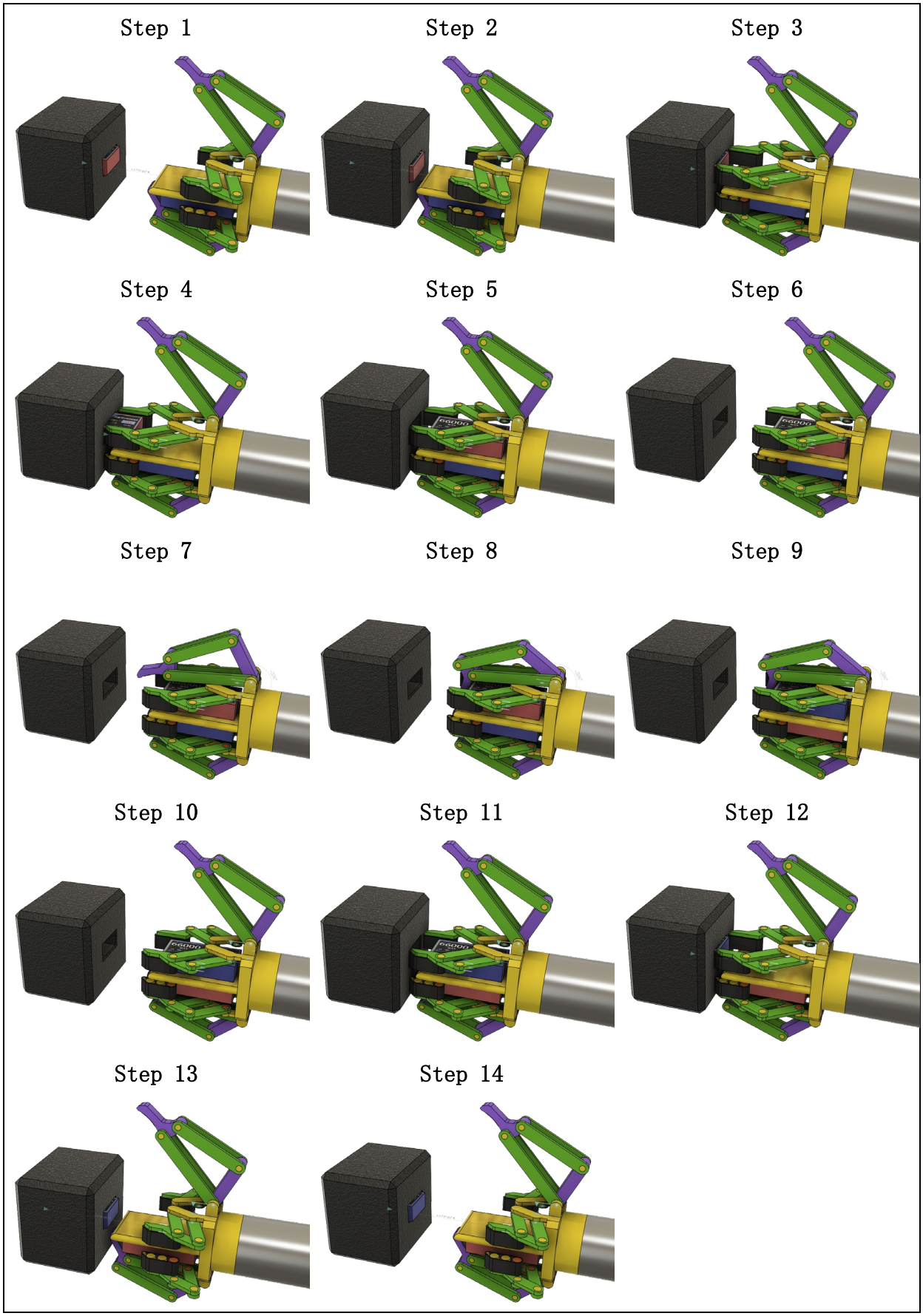


Fig. 2: The battery extraction/insertion sequence of Design A

2.2 Design B: Suction-Assisted Parallel Gripper

To address the challenge of gripping batteries that are deeply positioned in the drone's body (leaving only a small extracted part), Design B employs suction cup to pull the battery out slightly, increasing space for grabbing it. It consists of a standard two-finger parallel gripper with a suction cup mounted in the center as shown in the figure below.

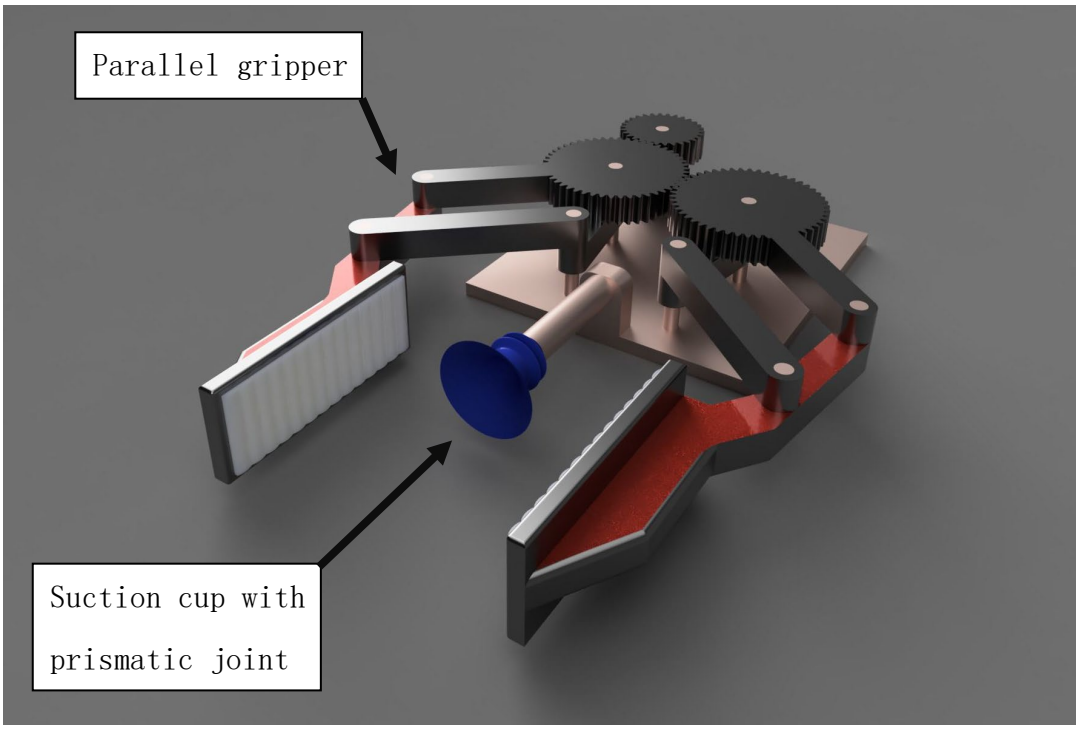


Fig. 3: Rendering of Design B (Suction-Assisted Parallel Gripper)

- Suction Engagement: The robot initially approaches the battery, and the suction cup activates to pull the battery outward by a few millimeters. This action increases the exposed length of the battery, making it much easier for the mechanical fingers to grip it securely.
- Parallel Finger Grasp: Once the battery is slightly exposed, the two parallel fingers close tightly around the battery body to secure it.
- Operational Sequence: Because this design only holds one battery, the robot must place the discharged battery at a designated storage location, then

move to a separate rack to pick up a charged battery, and finally insert it into the drone.

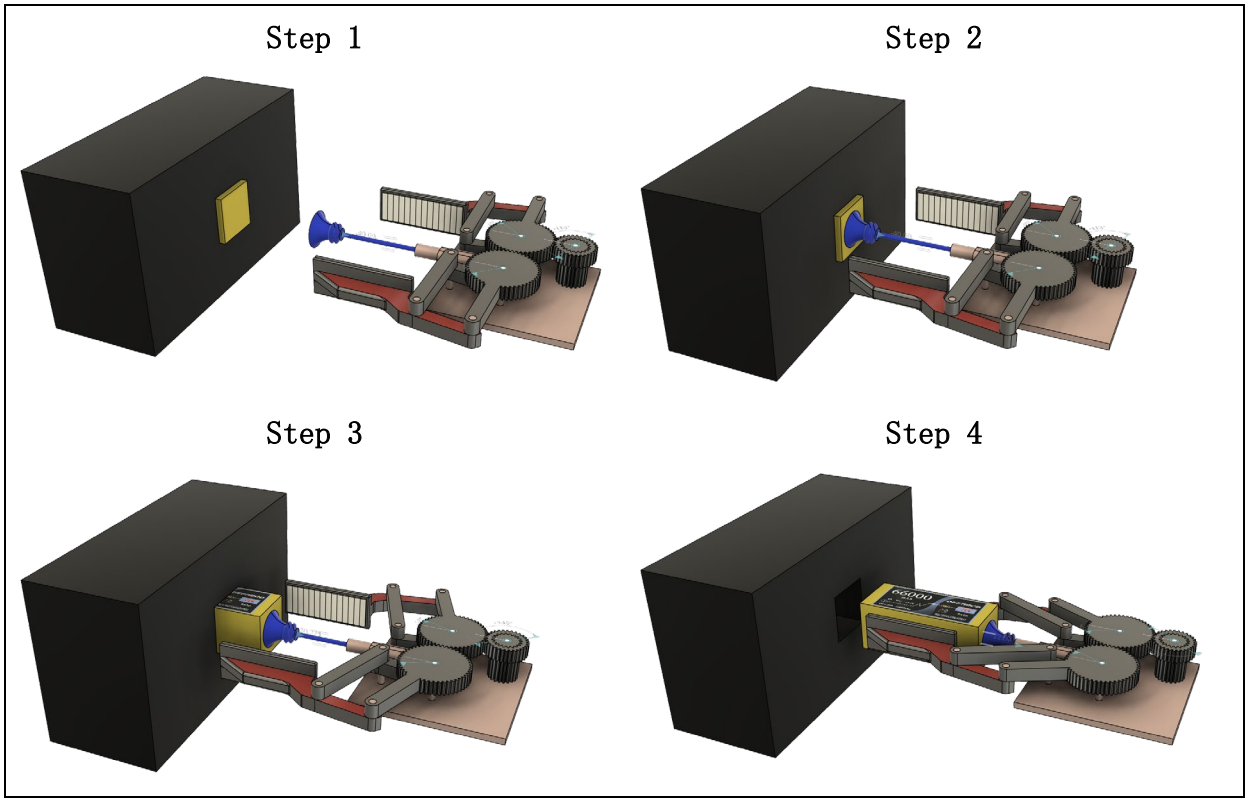


Fig. 4: Operational sequence of Design B

2.3 Simulation Framework (Isaac Lab)

The simulation replaces traditional Inverse Kinematics (IK) calculations with a learning-based approach. The environment was built in Isaac Lab using a 6-DOF UR10 manipulator with an authored parallel-jaw gripper.

The policy receives a 25-dimensional proprioceptive state: eight joint positions, eight joint velocities, the commanded end-effector pose (position + quaternion), and the finger state. It outputs eight joint position targets (six arm, two fingers). No IK solver is used anywhere in the control loop. Through reinforcement learning (RSL-RL PPO) and reward shaping, the robot learns to:

- ❖ Approach the battery face completely horizontally.
- ❖ Center the gripper on the battery, reaching a position error below 7 cm (from 1.05 m at start).

❖ Close the jaws once aligned.

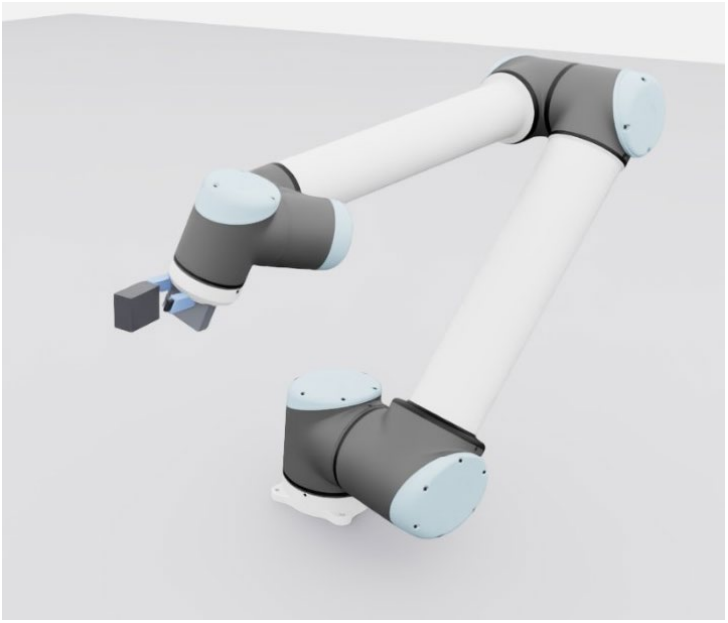


Fig. 5: UR10 robot learning battery grasping policy in Isaac Lab

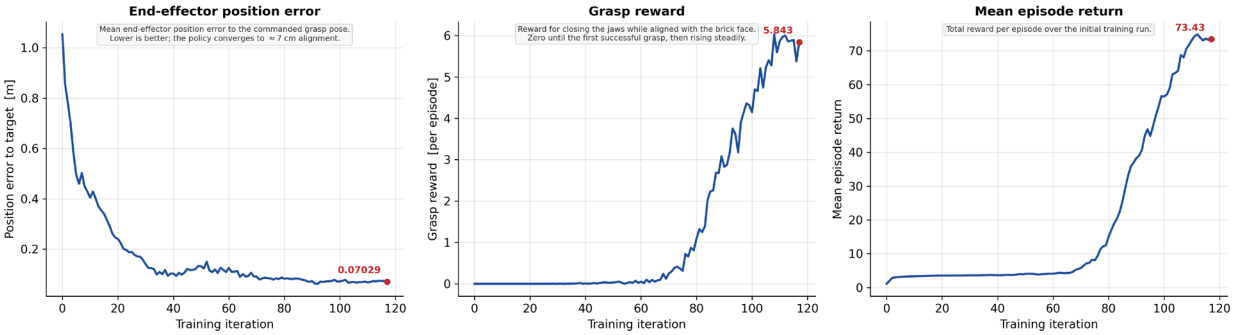


Fig. 6: Robot learning curves

Results and Conclusions

The study established a working simulation framework in which an RL policy replaces IK for the battery grasp-and-align task. The policy reduced end-effector alignment error from 1.05 m to below 7 cm and learned to close the jaws on the target (grasp reward: 0 → 6.3; mean return: +1.11 → +80.05). This confirms that the grasp task can be achieved without explicit IK solvers.

Design A has conceptually superior operational efficiency. By integrating dual batteries and the 180° roll ablility, the system can significantly reduce

the swap time and effectively bypass the problem of narrow battery tip grasping using its compliant belts.

Design B has a potential to be a highly effective solution. The integration of the central suction cup can successfully solve the problem of reaching and gripping deeply positioned batteries by pulling them outward, allowing the traditional parallel fingers to securely grasp the battery. However, it will require extra time to perform the place, pick, and insert sequence.

Future Work

Future steps include transferring the learned policies from Isaac Lab to the physical UR10 platform, extending the simulation to the full pick-and-insert sequence,simulating the two mechanical gripper designs, constructing physical prototypes of both grippers, and benchmarking their performance against industrial standards - measuring swap time, success rates, and alignment precision under the real-world conditions.

Appendix 1: Details from Simulation

```
Actor Model: MLPModel(
  (obs_normalizer): EmpiricalNormalization()
  (distribution): GaussianDistribution()
  (mlp): MLP(
    (0): Linear(in_features=25, out_features=256, bias=True)
    (1): ELU(alpha=1.0)
    (2): Linear(in_features=256, out_features=128, bias=True)
    (3): ELU(alpha=1.0)
    (4): Linear(in_features=128, out_features=64, bias=True)
    (5): ELU(alpha=1.0)
    (6): Linear(in_features=64, out_features=8, bias=True)
  )
```

```
)
Critic Model: MLPModel(
  (obs_normalizer): EmpiricalNormalization()
  (mlp): MLP(
    (0): Linear(in_features=25, out_features=256, bias=True)
    (1): ELU(alpha=1.0)
    (2): Linear(in_features=256, out_features=128, bias=True)
    (3): ELU(alpha=1.0)
    (4): Linear(in_features=128, out_features=64, bias=True)
    (5): ELU(alpha=1.0)
    (6): Linear(in_features=64, out_features=1, bias=True)
  )
)
--- Parameters ---
[INFO] Command Manager:  <CommandManager> contains 1 active terms.
+-----------------------------------+
|       Active Command Terms        |
+-------+---------+-----------------+
| Index | Name    |      Type       |
+-------+---------+-----------------+
|   0   | ee_pose | FixedPoseCommand |
+-------+---------+-----------------+

[INFO] Event Manager:  <EventManager> contains 1 active terms.
+-----------------------------------+
| Active Event Terms in Mode: 'reset' |
+---------+-------------------------+
|  Index  | Name                    |
+---------+-------------------------+
|    0    | reset_robot_joints      |
```

```
+---------+--------------------------+

[INFO] Recorder Manager:  <RecorderManager> contains 0 active terms.
+--------------------+
| Active Recorder Terms |
+-----------+---------+
|   Index   | Name    |
+-----------+---------+
+-----------+---------+

[INFO] Action Manager:  <ActionManager> contains 2 active terms.
+-----------------------------------+
|   Active Action Terms (shape: 8)   |
+-------+---------------+-----------+
| Index | Name          | Dimension |
+-------+---------------+-----------+
|   0   | arm_action    |        6 |
|   1   | gripper_action |        2 |
+-------+---------------+-----------+

[INFO] Observation Manager: <ObservationManager> contains 1 groups.
+-------------------------------------------------------+
| Active Observation Terms in Group: 'policy' (shape: (25,)) |
+------------+--------------------------+-------------+
|    Index    | Name                    |    Shape    |
+------------+--------------------------+-------------+
|     0     | joint_pos               |    (8,)    |
|     1     | joint_vel               |    (8,)    |
|     2     | pose_command            |    (7,)    |
|     3     | gripper_pos             |    (2,)    |
```

```
   +------------+-------------------------+------------+

   [INFO] Termination Manager:  <TerminationManager> contains 1 active
terms.
   +---------------------------+
   |   Active Termination Terms  |
   +-------+----------+----------+
   | Index | Name     | Time Out |
   +-------+----------+----------+
   |   0   | time_out |   True   |
   +-------+----------+----------+

   [INFO] Reward Manager:  <RewardManager> contains 7 active terms.
   +-------------------------------------------------+
   |               Active Reward Terms               |
   +-------+-----------------------------------+--------+
   | Index | Name                              | Weight |
   +-------+-----------------------------------+--------+
   |   0   | end_effector_keypoint_tracking    |   -0.5 |
   |   1   | end_effector_keypoint_tracking_exp |   1.5 |
   |   2   | action_rate                       | -0.001 |
   |   3   | action                            | -0.001 |
   |   4   | tcp_alignment                     |    0.5 |
   |   5   | grasp_success                     |    8.0 |
   |   6   | gripper_action                    |  -0.02 |
   +-------+-----------------------------------+--------+

   [INFO] Curriculum Manager:  <CurriculumManager> contains 0 active
terms.
   +--------------------+
```

Active Curriculum Terms

Index	Name